REQUIREMENTS GATHERING

1. Stakeholder Analysis

Stakeholder	Needs and Expectations
Deaf and Hard-of-Hearing Users	An AI-Powered mobile application that recognizes sign language gestures in real time and displays the corresponding Arabic words.
People Who Do Not Understand Sign Language	A simple way to understand sign language gestures through Arabic text.
AI/ML Developer	Suitable sign language datasets, accurate gesture recognition, and successful integration of the AI model with the application.
Mobile Application Developer	A responsive and user-friendly mobile application integrated with the AI model through an online API.
UI/UX Designers	A simple, intuitive, and accessible interface that makes live translation easy to use.

2. User Stories & Use Cases

2.1 User Stories

- As a **deaf user**, I want to open the camera and sign **so that** my signs are converted to Arabic text in real time.
- As a **deaf user**, I want the app to show the recognized word immediately **so that** I can communicate in a live conversation.
- As a **hearing person**, I want to read the translated text on the screen **so that** I can understand the person signing.
- As a **user**, I want clear feedback when the sign isn't recognized **so that** I can repeat it.
- As a **user**, I want simple Arabic UI **so that** I can use the app without any training.

2.2 Use Cases

Use Case 1: Start Sign Recognition

Field	Description
Actor	Deaf User
Precondition	The application is open.
Trigger	The user opens the camera screen.
Main Flow	1. The user opens the Sign Recognition screen. 2. The application requests camera access if needed. 3. The camera starts capturing video.
Postcondition	The camera is ready to capture sign language gestures.

Use Case 2: Perform Sign Gesture

Field	Description
Actor	Deaf User
Precondition	The camera is active and ready.
Trigger	The user performs a sign gesture in front of the camera.
Main Flow	1. The user performs an Arabic sign. 2. The camera captures the sign. 3. The captured input is prepared for processing.
Postcondition	The sign input is ready for feature extraction.

Use Case 3: Extract Sign Features

Field	Description
Actor	Sign Recognition System
Precondition	Sign video input is available.
Trigger	The system receives captured video frames.
Main Flow	1. The system identifies hand, face, and body landmarks. 2. The landmarks are extracted. 3. The extracted landmarks are organized into structured features.
Postcondition	The extracted features are ready for preprocessing and prediction.

Use Case 4: Predict Sign

Field	Description
Actor	Trained ML Model
Precondition	The extracted features are preprocessed and ready.
Trigger	The system sends the prepared features to the trained model.
Main Flow	1. The model receives the input features. 2. The model analyzes the sign patterns. 3. The model predicts the corresponding sign class. 4. The predicted class is mapped to its Arabic word.
Postcondition	The predicted Arabic word is returned to the application.

Use Case 5: Display Prediction

Field	Description
Actor	Deaf User
Precondition	The system has received a prediction result.
Trigger	The recognition process returns an Arabic word.
Main Flow	1. The application receives the predicted word. 2. The application displays the Arabic word on the screen.
Postcondition	The user can see the recognized Arabic word.

3. Functional Requirements

ID	Functional Requirement
FR-01	The system shall provide a mobile application interface for accessing its main features.
FR-02	The system shall capture sign language gestures in real time.
FR-03	The system shall process the captured input using the sign language recognition AI model.
FR-04	The system shall communicate with the AI recognition service through an API or server, according to the implemented architecture.
FR-05	The AI model shall recognize supported Arabic sign language gestures and predict their corresponding words.
FR-06	The system shall display recognized words in Arabic on the mobile application's screen.
FR-07	The system shall continue live recognition while the translation feature is active.
FR-08	The system shall handle unrecognized gestures, network failures, and service errors appropriately.
FR-09	The system shall provide a Settings screen for the preferences supported by the application.

4. Non-functional Requirements

Category	Non-functional Requirement
Performance	The system should recognize gestures and display results with minimal delay to support real-time interaction.
Accuracy	The AI model should achieve an acceptable recognition accuracy for the supported sign language vocabulary, measured through testing.
Usability	The application should provide a clear, simple, and easy-to-navigate user interface.
Accessibility	Arabic text output should be readable, clearly presented, and easy to understand.
Security	Communication between the mobile application and the online recognition service should use HTTPS.
Privacy	Camera input and related data should be handled responsibly and should not be stored unnecessarily.
Reliability	The application should handle recognition errors, network interruptions, and API failures without unexpected crashes.
Compatibility	The application should operate on the Android devices and operating system versions supported by the project.
Maintainability	The mobile application, API, and AI model should be structured to facilitate testing, maintenance, and future improvements.
Availability	The online recognition service should be accessible when the service is operational and the user has an internet connection.

5. Project Scope and Limitations

In Scope

- Real-time recognition of supported Arabic sign language gestures.
- Translation from sign language gestures into Arabic written words.
- Live camera input through the mobile application.
- Integration between the mobile application and the AI recognition service.
- Displaying recognition results and handling errors.

Out of Scope

- Translating Arabic written text into sign language.
- Supporting languages other than Arabic.
- Saving and viewing translation history.
- Translating complete sentences unless this functionality is explicitly implemented and supported by the AI model.